

Independent Vector Analysis Applied to Remove Muscle Artifacts in EEG Data

Xun Chen, *Member, IEEE*, Hu Peng, Fengqiong Yu, and Kai Wang

Abstract—Electroencephalogram (EEG) data are often contaminated by various electrophysiological artifacts. Among all these artifacts, the muscle activity is particularly difficult to remove. In the literature, independent component analysis (ICA) and canonical correlation analysis (CCA), as blind source separation techniques, are the most popular methods. In this paper, we introduce a novel method for removing muscle artifacts in EEG data based on independent vector analysis. This method exploits both the second-order and higher order statistical information and thus takes advantage of both ICA and CCA. The proposed method is evaluated on realistic simulated data and is shown to significantly outperform ICA and CCA. In addition, the proposed method is applied on real ictal EEG data seriously contaminated with muscle artifacts. The proposed method is able to largely suppress muscle artifacts without altering the underlying EEG activity.

Index Terms—Blind source separation (BSS), electroencephalogram (EEG), independent vector analysis (IVA), muscle artifacts.

I. INTRODUCTION

ELECTROENCEPHALOGRAPH (EEG) is a popular measurement modality for recording the electrical activity generated by populations of neurons of the cerebral cortex. It has been widely used in a variety of practical applications. For instance, Liang *et al.* [1] and Lay-Ekuakille *et al.* [2] employed EEG to monitor the status of human sleep. Wang and Zhang [3] used motor imagery EEG to identify human movement intention in brain-computer interface. Dai *et al.* [4] exploited EEG as a biometrics for person identification. Chen *et al.* [5] utilized EEG combined with kinematic data to study pathological brain mechanisms for movement disorder. In the meantime, novel EEG acquisition instruments have also been designed to measure high-quality EEG signals even in mobile wearable situations [6], [7]. Nevertheless, EEG is often contaminated by various noncerebral electrophysiological artifacts, making practical applications problematic.

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X. Chen and H. Peng are with the Department of Biomedical Engineering, Hefei University of Technology, Hefei 230009, China (e-mail: xunchen@ece.ubc.ca; hpeng@hfut.edu.cn).

F. Yu and K. Wang are with the Department of Neurology, the First Affiliated Hospital of Anhui Medical University, Hefei 230022, China.

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Therefore, signal denoising, as one of the important topics in the instrumentation and measurement (I&M) field, is crucially necessary for EEG processing.

It is widely acknowledged that, among these artifacts, muscle artifacts induced by muscle contraction (e.g., biting, chewing, and frowning) are particularly difficult to remove [8], [9]. As already reported in [9], the unique challenges associated with muscle artifact removal are especially due to their high amplitude, large frequency range, and variable topographical distribution. Analogous to denoising other types of signals [10]–[13] in the I&M field, blind source separation (BSS) techniques, including independent component analysis (ICA) and canonical correlation analysis (CCA), were mostly utilized [14], [20].

ICA, as a well-known BSS technique, has been widely used to remove muscle artifacts in EEG [14], [15]. In most ICA algorithms, multichannel EEG data are linearly unmixed into statistically ICs by exploiting higher order statistics (HOS). The derived ICs, representing the underlying sources, can be inspected and those that resemble muscle artifacts can be removed during reconstruction, leaving clean data. While many researchers recommend ICA [15]–[17], disagreement exists in the literature about the effectiveness of ICA in removing muscle artifacts from EEG data [9], [18]. It has been reported that most ICs actually contain both brain and myogenic activity [19]. By exploiting HOS, ICA is quite capable of isolating the artifacts with stereotyped scalp topographies into one single IC. However, the shapes and amplitudes of the artifacts depend on the degree of muscle contraction, on the type of muscles contracted and on the number of muscles involved. Thus, they are hard to stereotype.

Recently, CCA has been proposed as an alternative BSS technique for muscle artifact removal [20], [21]. Using original EEG as the first data set and its time-delayed version as the second data set, CCA aims to extract the sources maximally autocorrelated and meanwhile mutually uncorrelated by exploiting the second-order statistics (SOS). In comparison with brain activity, muscle artifacts have relatively low autocorrelation due to their broad frequency spectrum. CCA utilizes this distinguishable feature to isolate muscle artifacts from brain activity. Clercq *et al.* [20], Gao *et al.* [21], and Mowla *et al.* [22] have demonstrated that the CCA outperforms ICA on synthetic data. Clinical applications also support the usage of CCA due to improved performance [23], [24]. The results have been further documented by Chen *et al.* [26] when dealing with the muscle artifact removal problem in single-channel EEG.

However, as reported in a very recent survey [15], which method performs the best is highly dependent on the specific EEG data at hand. While CCA utilizes SOS to explore the autocorrelation characteristic with the recovered sources only statistically uncorrelated, ICA exploits HOS to explore the statistical independence. In statistics, independence is a much stronger condition than uncorrelatedness and, while the former implies the latter, the converse is not necessarily true. Therefore, for muscle artifact removal, both CCA and ICA may have their individual advantages and disadvantages, and fortunately, perhaps most importantly, these two methods can be considered complementary from their exploited statistical information.

In this paper, we propose to integrate the advantages of CCA and ICA into one single BSS framework. In other words, we attempt to simultaneously exploit SOS and HOS and thus solve this denoising problem better. Independent vector analysis (IVA) [28], a recently emerging BSS technique, is a method of this kind. IVA can be treated as an extension of ICA from one to multiple data sets. It employs criteria related to information theory to decompose each data set into mutually independent sources, and meanwhile utilizes the dependence information between data sets to make the corresponding sources across data sets dependent with each other. When modeling the dependence using SOS, the corresponding sources are expected to be maximally correlated, which is exactly the advantage of CCA for solving the denoising problem. Therefore, IVA is a natural way to combine the advantages of both methods. It is anticipated that IVA will achieve better performance than either of the two. We will examine the performance of the proposed IVA method and compare it with the most commonly used ICA and CCA methods. We first validate the methods on realistic simulated data. We then apply them to challenging real-life EEG recordings contaminated with muscle artifacts.

The main contribution of this paper is summarized as follows. To the best of our knowledge, it is the first time in the literature that both HOS and SOS are considered under the IVA framework for muscle artifact removal from multichannel EEG data. The direct merit of using IVA is that it can help to significantly reduce uncertainties in the measurement procedures, thus relaxing the instrument front-end design constraints, enhancing the reliability of instrument systems, and improving the accuracy of subsequent signal analysis. While the IVA-based method is intentionally proposed for decomposing EEG signals in this paper, it is worth noting that it can also be applied to analyze other types of signals in the I&M field, including, but not limited to, thermography [10], X-ray image [11], electrocardiography [12], power signals [29], partial discharges [30], [31], and photoplethysmograph [32], especially, the decomposition of which may need HOS and SOS simultaneously. That is to say, if the noise to be isolated not only possesses relatively unique morphological distribution, which requires HOS, but also exhibits relatively random property, which also needs the second-order autocorrelation for assistance, IVA is a superior candidate. A more theoretical analysis from the statistical diversity point of view will be presented in Section IV.

The remainder of this paper is organized as follows. In Section II, the material and the methodology are described, including the information for EEG data and the BSS techniques used. In Section III, the comparison results by applying the methods to both synthetic and real data are reported. Discussion and conclusion are presented in Section IV.

II. MATERIAL AND METHODOLOGY

Notations: Scalars are denoted by lowercase italic letters ($a, b, \dots$), vectors by lowercase boldface letters ($\mathbf{a}, \mathbf{b}, \dots$), matrices by boldface capitals ($\mathbf{A}, \mathbf{B}, \dots$), and the number of rows and columns by italic capitals ($P, T, \dots$). A matrix or vector transposition is denoted by an uppercase superscript T (e.g., $\mathbf{X}^T$ and $\mathbf{v}^T$).

A. Data Description

1) Realistic Simulated Data:

a) Brain activity: Eighteen healthy subjects (ten males and eight females with the mean age 29.3) were recruited for pure EEG data collection. They were instructed to keep their eyes close and their bodies relax to avoid eye blink/movement and muscle artifacts. The EEG recordings were collected using an EEG Quick-Cap and a NuAmps amplifier (Compumedics Neuroscan, El Paso, TX) with a bandpass filtering between 1 and 70 Hz on 19 channels (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, and O2) according to the International 10–20 System. The sampling frequency was 1000 Hz and an average reference was adopted. Through the visual inspection of an independent experienced neurophysiologist, a 10-s EEG epoch with no muscle artifact was selected from each subject. Thus, there are 18 pure EEG data matrices with size 19×10000 in the subsequent analysis, denoted as $\mathbf{X}_{\text{EEG}}^{(i)}$'s ($i = 1, 2, \dots, 18$).

b) Muscle activity: Referring to previous studies [20]–[22], the second-order blind identification [34] was used to extract pure muscle artifacts from contaminated EEG data. A neurophysiologist labeled muscle artifacts from all of the components in the decomposition by inspecting some features, such as the power spectral density (PSD), time series, and topography. It is worth emphasizing that, for a large number of components, no clear separation between muscle and brain activity was established. However, in some cases, a component containing pure muscle activity could be observed. By keeping this component and replacing all other components with zeros, a conventional EEG format, only containing the effects of the identified muscle activity to each channel, would be reconstructed by multiplying the processed components with the corresponding field distribution. This procedure was repeated for four different 10-s EEG epochs from four different subjects. Note that the four extracted components represent different types of muscle activities, including chewing, mouth opening, and head tilting. The averaged summation of these four independent muscle activities with the conventional EEG format was stored in a 19×10000 dimensional matrix $\mathbf{X}_{\text{muscle}}$.

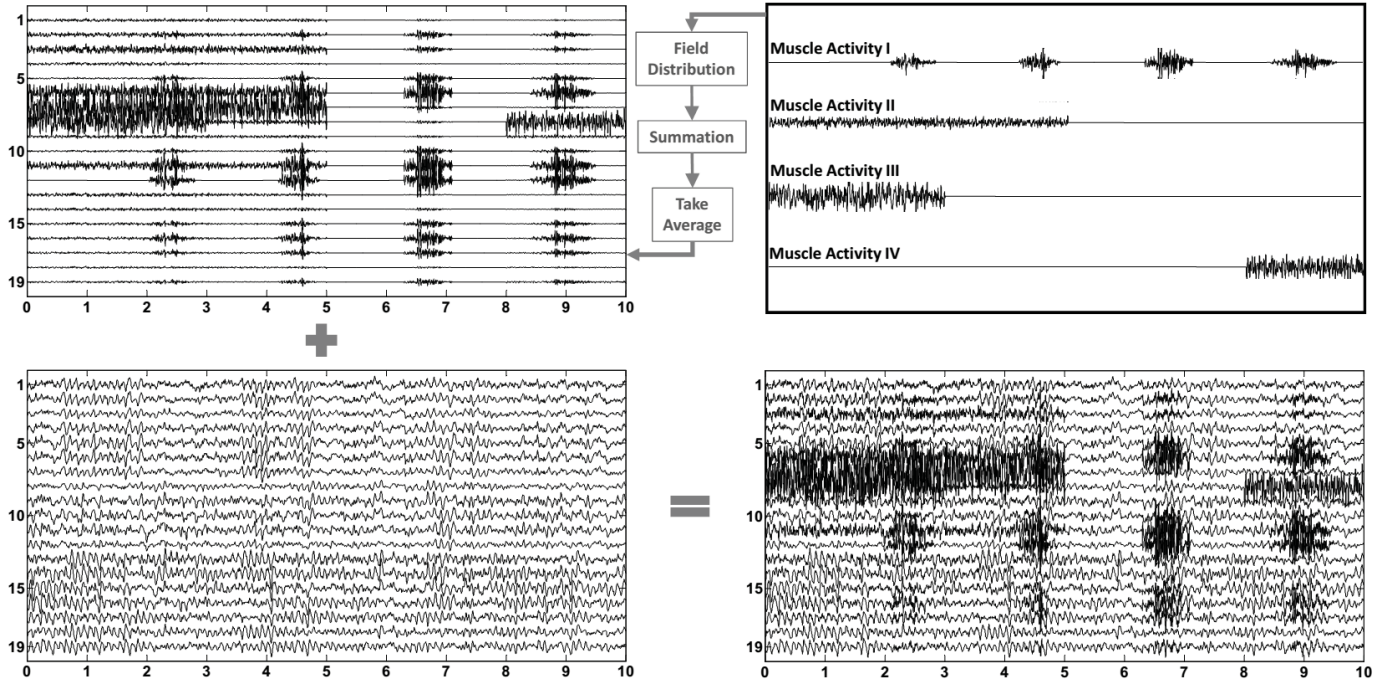


Fig. 1. Schematic of generating realistic simulated data. The horizontal axis denotes time with the unit second and the vertical axis denotes channel index with Channels 1–19 representing Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, and O2, respectively. The pure EEG data are from one subject and the SNR value here is one.

c) *Mixed data*: The mixed data were generated by superimposing the summed-up muscle activity on the pure EEG data as follows:

$$\mathbf{X}_{\text{mixed}}^{(i)} = \mathbf{X}_{\text{EEG}}^{(i)} + \lambda \cdot \mathbf{X}_{\text{muscle}}, \quad i = 1, 2, \dots, 18 \quad (1)$$

with λ indicating the contribution of muscle activity. Through changing the parameter λ , signal-to-noise ratio (SNR) can be adjusted as follows:

$$\text{SNR} = \frac{\text{RMS}(\mathbf{X}_{\text{EEG}}^{(i)})}{\text{RMS}(\lambda \cdot \mathbf{X}_{\text{muscle}})} \quad (2)$$

where the root-mean-squared (rms) value is defined as

$$\text{RMS}(\mathbf{X}) = \sqrt{\frac{1}{P \cdot T} \sum_{p=1}^P \sum_{t=1}^T \mathbf{X}^2(p, t)} \quad (3)$$

with P representing the number of EEG channels and T denoting the number of temporal samples. Simulations were performed on the 18 mixed data with different SNR values. Fig. 1 shows the specific procedure for generating an example mixed data with SNR = 1.

2) *Real Ictal Data*: In Fig. 2, a 10-s 21-channel scalp EEG recording from a long-term epilepsy monitoring unit, Onafhankelijke Software Groep EEG recorders (Rumst, Belgium), is shown. This recording contains ictal activity from a patient with mesial temporal lobe epilepsy, severely contaminated with eye blinks and muscle artifact. The sampling frequency was 250 Hz. The EEG was bandpass filtered between 0.3 and 35 Hz. Eye artifacts can be observed around 2.5, 3.5, 6, and 7.5 s, and are most observed in the Fp1 and Fp2 channels. Seizure activity is constantly present in the T2, F8, T4, and

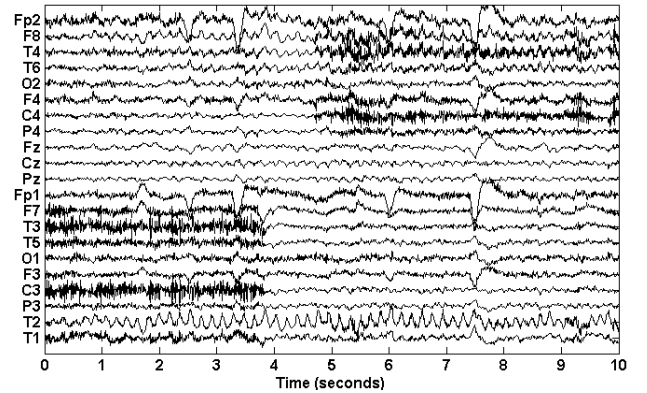


Fig. 2. Ten-second ictal scalp EEG recordings contaminated with eye blinks and muscle artifacts.

T6 channels. Muscle activity starts on one side of the head, which could be observed between 0 and 3.9 s on channels, such as F7 and T3, and between 5 and 10 s on channels, such as F8 and T4. Ictal EEG is often severely contaminated with muscle artifacts, making the determination and localization of the ictal onset complicated. Successfully separating muscle activity from this ictal data is particularly challenging. We will demonstrate this point by comparing the decomposition results of the methods.

B. Methods

In the following method description, multichannel EEG data contaminated with artifacts are denoted by a matrix $\mathbf{X}$ with size $P \times T$, where P and T have the same meaning as defined in (3). It can also be expressed by a time course

$\mathbf{X}(t) = [\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_P(t)]^T$ ($t = 1, 2, \dots, T$). It is assumed to be the linear mixture of P unknown source signals $\mathbf{S}(t) = [\mathbf{s}_1(t), \mathbf{s}_2(t), \dots, \mathbf{s}_P(t)]^T$. The matrix multiplication can be expressed as

$$\mathbf{X}(t) = \mathbf{A}\mathbf{S}(t) \quad (4)$$

where $\mathbf{A}$ is the mixing matrix with size $P \times P$. It can also be transformed as

$$\tilde{\mathbf{S}}(t) = \mathbf{W}\mathbf{X}(t) \quad (5)$$

where $\tilde{\mathbf{S}}(t)$ represents the recovered sources and $\mathbf{W}$ is the unmixing matrix, ideally, the inverse of the mixing matrix $\mathbf{A}$. To obtain artifact-free EEG data, the sources related to muscle activity are identified and removed during reconstruction. The ultimate goal is to figure out the unmixing matrix $\mathbf{W}$ and recover the original sources only based on the observations $\mathbf{X}(t)$. Unless there are extra constraints imposed, it is, in general, impossible to solve the source separation problem. Therefore, the key issue here is what types of statistics should be exploited to build specific cost functions in the BSS model to fit and solve the problem better.

1) *ICA*: ICA unmixes linearly mixed signals using only recorded time courses by imposing statistical independence of the sources. ICA comprises several different implementations. As it is not in the scope of this paper to investigate all different ICA algorithms, the representative joint approximate diagonalization of eigenmatrices algorithm [35], exploiting the fourth-order cumulants (HOS), is adopted to decompose the contaminated EEG data, given that it was used in several previous studies for muscle artifact removal in scalp EEG recordings [8], [20].

2) *CCA*: CCA solves the problem using the recorded EEG data and its time-delayed version by maximizing the correlation coefficient (SOS) between the corresponding sources from the two data sets [36]. In this case, the sources extracted from the EEG recordings are forced to be maximally autocorrelated and mutually uncorrelated. In other words, the sources are sorted in terms of their autocorrelation. Due to the relatively lower autocorrelation, muscle artifacts could be isolated into the last several sources using CCA. Note that as both ICA and CCA are well-known BSS techniques in the literature, we only briefly describe their basic concepts. For more details, one can refer to the related ICA works [14]–[16] and CCA ones [20]–[27].

3) *IVA*: IVA is a generalization of ICA from one to multiple data sets, originally designed to address the permutation problem in the frequency domain for the separation of acoustic sources [37]. In [28], IVA was formulated as a general joint BSS framework to ensure that the corresponding sources extracted from different data sets are maximally dependent while the sources within each data set are independent of each other. That is to say, source independence within one data set and corresponding source dependence across multiple data sets are maximized simultaneously. In particular, in IVA, the concept of source component vector (SCV) was defined across multiple data sets [37]. The p th SCV, $\mathbf{s}_p = [s_p^{[1]}, s_p^{[2]}, \dots, s_p^{[M]}]^T$ ($p = 1, 2, \dots, P$), is a random vector independent of all other SCVs and the components within

each SCV are dependent. The symbol $s_p^{[m]}$ ($m = 1, 2, \dots, M$) represents the p th source component in the m th data set. The goal of IVA is to identify the independent SCVs from multiple multidimensional data sets, achieved by minimizing the mutual information among SCV estimates $\hat{\mathbf{s}}_p$

$$\begin{aligned} I_{\text{IVA}} &\triangleq I[\hat{\mathbf{s}}_1; \hat{\mathbf{s}}_2; \dots; \hat{\mathbf{s}}_P] \\ &= \sum_{p=1}^P H[\hat{\mathbf{s}}_p] - H[\hat{\mathbf{s}}_1, \hat{\mathbf{s}}_2, \dots, \hat{\mathbf{s}}_P] \\ &= \sum_{p=1}^P H[\hat{\mathbf{s}}_p] - H[\mathbf{W}^{[1]}\mathbf{x}^{[1]}, \dots, \mathbf{W}^{[M]}\mathbf{x}^{[M]}] \\ &= \sum_{p=1}^P \left(\sum_{m=1}^M H[\hat{\mathbf{s}}_p^{[m]}] - I[\hat{\mathbf{s}}_p] \right) \\ &\quad - \sum_{m=1}^M \log|\det(\mathbf{W}^{[m]})| - C \end{aligned} \quad (6)$$

where H is the entropy, $I[\hat{\mathbf{s}}_p]$ is the mutual information within the p th SCV, $\mathbf{W}^{[m]}$ is the unmixing matrix for the m th data set, and C is the constant term $H[\mathbf{x}^{[1]}, \dots, \mathbf{x}^{[M]}]$ only depending on the observations. From the final representation of (6), minimizing the IVA cost function is equivalent to minimizing the entropy of all components $\hat{\mathbf{s}}_p^{[m]}$, and meanwhile maximizing the mutual information within each estimated SCV $\hat{\mathbf{s}}_p$. The former leads to the maximization of source independence within each data set, and the latter results in the maximization of source dependence across multiple data sets.

This cannot be achieved by running separate ICA on each of the data sets, since ICA only focuses on source independence within individual data sets, but without paying any attention to source dependence across multiple ones. Likewise, this cannot be achieved by performing CCA on the data sets with $M = 2$ either, in that CCA only exploits the SOS to ensure source dependence across data sets and source uncorrelatedness within each individuals, but without any guarantee for source independence within each data set. Therefore, IVA is a natural way to integrate ICA and CCA. In other words, IVA is a natural generalization of both methods to the case where both SOS and HOS are considered.

As the implementation algorithms involve the selection of specific probability distributions for the SCVs, IVA comprises several related methods. The most widely used ones include IVA-L [37], which assumes that each SCV follows a multivariate Laplace distribution that is isotropic and possesses no second-order correlation, and IVA-G [28], which assumes each SCV is multivariate Gaussian distributed. Recently, IVA-GL, using the IVA-G solution to initialize the IVA-L algorithm, has been recommended, since it has been shown that this yields a more robust estimator than using either IVA-L or IVA-G [28], [38]. The implementation first considers full second-order dependence among entries of an SCV by IVA-G. The estimates of the demixing matrices are then employed to initialize IVA-L, and HOS is considered by assuming a Laplacian distribution for each entry within an SCV. In this paper, we will utilize IVA-GL by considering the importance of the second-order and higher order information, which can

simultaneously exploit the temporal and spatial structure of muscular artifacts. Using the recorded EEG data as the first data set and its time-delayed version as the second one, it is anticipated that IVA-GL can decompose the recorded EEG data into the self-correlated and mutually independent sources sorted in terms of their autocorrelation. In this case, muscle artifacts should be isolated into the last several independent sources. Removing them during reconstruction leads to cleaned EEG data.

Regarding the computational complexity, to the best of our knowledge, there is no published study systematically discussing this issue. In this paper, we used the IVA-G and IVA-L that use the decoupling trick. For IVA-G, the covariance matrices have to be estimated for each of the SCVs. For IVA-L, this estimation does not have to be performed, because in the algorithm, it is assumed that the data are uncorrelated, so that the scatter or dispersion matrix of each SCV is the identity. Thus, regarding IVA-L, the computational complexity for separating one SCV is $O(TPM)$. Therefore, the computational complexity for each full iteration is $O(P^2TM)$. For IVA-G, it is $O(P^2TM + PM^2T)$.

4) *Performance Measures*: To quantitatively examine the performance of the methods, three objective measures were adopted for the simulated data. The first measure was the relative rms error (RRMSE) in the temporal domain, termed as $RRMSE_{\text{temporal}}$. It is defined as follows:

$$RRMSE_{\text{temporal}}^{(i)} = \frac{\text{RMS}(\mathbf{X}_{\text{EEG}}^{(i)} - \tilde{\mathbf{X}}_{\text{EEG}}^{(i)})}{\text{RMS}(\mathbf{X}_{\text{EEG}}^{(i)})} \quad (7)$$

with $\tilde{\mathbf{X}}_{\text{EEG}}^{(i)}$ denoting the i th reconstructed EEG data after muscle artifact removal. The second measure was the RRMSE in the spectral domain using PSD, denoted as $RRMSE_{\text{spectral}}$. It is defined as the following:

$$RRMSE_{\text{spectral}}^{(i)} = \frac{\text{RMS}(\text{PSD}_{\text{EEG}}^{(i)} - \tilde{\text{PSD}}_{\text{EEG}}^{(i)})}{\text{RMS}(\text{PSD}_{\text{EEG}}^{(i)})} \quad (8)$$

with $\text{PSD}_{\text{EEG}}^{(i)}$ denoting the PSD of the i th original pure EEG and $\tilde{\text{PSD}}_{\text{EEG}}^{(i)}$ denoting the PSD of the i th reconstructed EEG after muscle artifact removal. Here, the frequency band used in PSD was 1–70 Hz, since EEG was preprocessed by 1–70-Hz bandpass filter as described in Section II-A1. To further evaluate the capability of preserving the original EEG changing patterns, correlation coefficient between the original EEG in each channel and its reconstructed counterpart was also calculated. These correlation coefficient values were then used to compute the average correlation coefficient $[\text{ACC}^{(i)}]$ over all channels of the i th EEG data, which was treated as the third evaluation measure. Finally, the average $RRMSE_{\text{temporal}}$, $RRMSE_{\text{spectral}}$, and ACC with the corresponding standard deviations over all 18 data matrices at each SNR value were calculated to demonstrate the performance of the methods. For the real ictal EEG, qualitative analysis would be provided, since the ground truth was unknown.

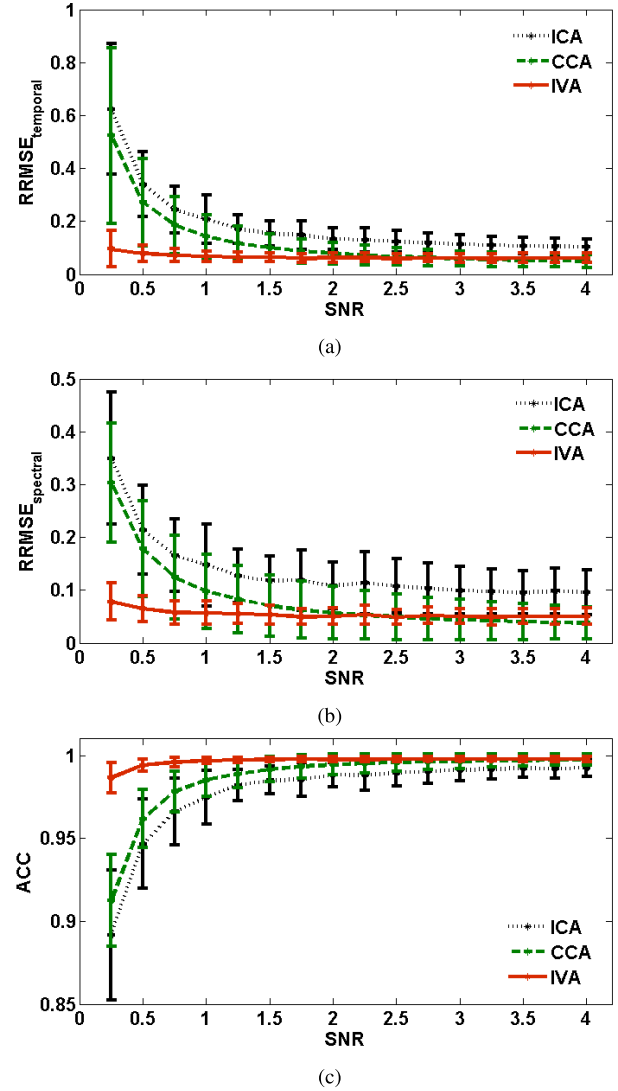


Fig. 3. Simulation study. The performance comparisons of the three BSS techniques at various SNR values in terms of three measures. (a) $RRMSE_{\text{temporal}}$. (b) $RRMSE_{\text{spectral}}$. (c) ACC.

III. RESULTS

A. Results of Simulation Study

In this section, we applied ICA, CCA, and IVA to the simulated 19-channel EEG data for muscle artifact removal, and performed a quantitative comparison. We examined the reliability of the three methods at different SNR values in terms of $RRMSE_{\text{temporal}}$, $RRMSE_{\text{spectral}}$, and ACC, as mentioned in Section II-B4. Remember that we have a total of 18 pure EEG data sets as described in Section II-A1. At each SNR value, each method was performed 18 times independently on the 18 mixed data sets. In each run, 19 underlying sources were generated and the cleaned EEG data were obtained by setting the sources of muscle activity to zero during reconstruction. The average $RRMSE_{\text{temporal}}$, $RRMSE_{\text{spectral}}$, and ACC values with standard deviations at each SNR value were calculated as the final evaluation measures.

The performance of ICA, CCA, and IVA is shown in Fig. 3. It is observed that IVA consistently yields superior performance at different SNR values in terms of $RRMSE_{\text{temporal}}$ in

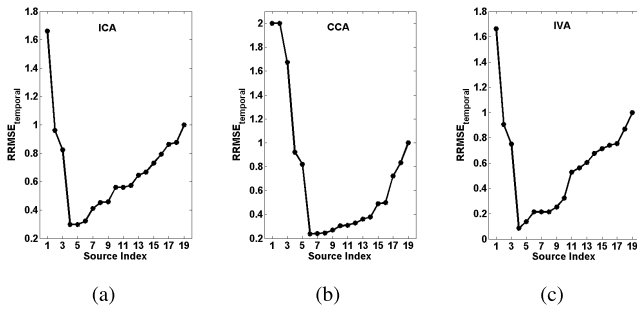


Fig. 4. $RRMSE_{temporal}$ versus the number of removed sources at $SNR = 0.5$ for one EEG data set using the three methods.

Fig. 3(a), $RRMSE_{spectral}$ in Fig. 3(b), and ACC in Fig. 3(c). When EEG data are lightly or moderately contaminated (e.g., $SNR > 2$), the $RRMSE_{temporal}$, $RRMSE_{spectral}$, and ACC values of the three methods are quite close. In particular, the performance of CCA and IVA is almost identical. However, as the contamination level increases (e.g., $SNR < 2$), the performance of ICA and CCA degrades sharply compared with that of IVA. Both ICA and CCA seems incapable to preserve the original EEG signals as well as IVA does. The results suggest that IVA, as a novel BSS technique, can be a better candidate than the traditional ICA and CCA for denoising muscle artifact from multichannel EEG data.

It should be noted that, for all the three methods, the extracted sources are sorted according to their autocorrelations from high to low. Obviously, CCA and IVA can automatically do this, since they both use SOS to explore the dependence information between original data and the time-delayed version. They priorly, gradually extract sources with relatively higher autocorrelations. Regarding ICA, the order of extracted sources is originally uncertain. However, it is apparent that the sources, containing only muscle activity or mostly, possess the characteristic of quite low autocorrelations. Even the sources mixed by brain and muscle activity also have the similar property. Hence, it is likely reasonable in a follow-up procedure to sort the ICA sources in terms of autocorrelations from high to low for further denoising processing. The purpose of all the above-mentioned efforts is to make the quantitative comparison more convenient and fairer as described in the following.

With the well-ordering decomposition results, muscle artifacts can be readily identified from the last several sources due to their low autocorrelations. Since the ground truth is known in this simulated study, the best performance of the three methods at each SNR value is presented by selecting and removing an optimal number of sources during reconstruction. Here, the concept of optimal number means that, by removing this number of the last sources, the corresponding RRMSE value should be the lowest in comparison to the other numbers. Fig. 4 shows an example of the RRMSE value versus the number of removed sources by applying ICA, CCA, and IVA to one contaminated 19-channel EEG data set with $SNR = 0.5$, as shown in Fig. 5. The lowest RRMSE values in the three cases happened at the numbers 4, 6, and 4, respectively. Hence, for instance, the optimal result of IVA was obtained when the last four sources were removed.

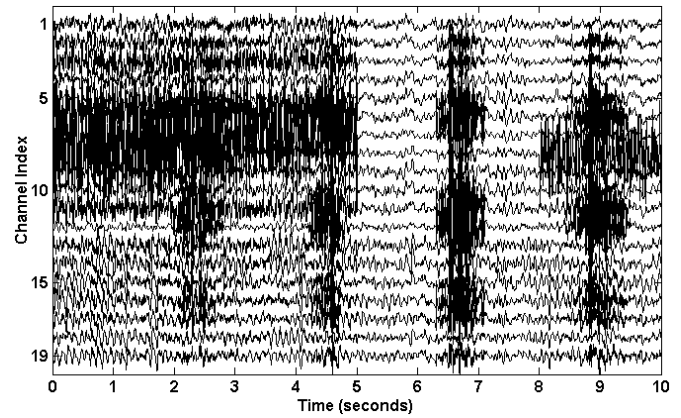


Fig. 5. Example contaminated 19-channel EEG data set with $SNR = 0.5$.

To better demonstrate the differences among these methods, some detailed results are also provided, as shown in Fig. 6. Fig. 6(a)–(c) presents the sources extracted by applying ICA, CCA, and IVA, respectively, to the severely contaminated EEG data set in Fig. 5. Here, the normalized sources are presented to examine their details more clearly. Roughly speaking, all the three methods are able to isolate muscle artifacts into the last several sources, e.g., the last four sources in ICA and IVA. Note that the last two sources in CCA are not muscle artifacts, but actually very tiny random noise with extremely low amplitude (e.g., 10^{-8}) so that removing these two sources during reconstruction has no impact on the RRMSE value, which can be seen from Fig. 4(b). Thus, we only need to pay close attention to the remaining four sources of the last six in CCA.

By ICA, the last four sources have quite similar changing patterns to the original muscle activity shown in Fig. 1. However, it will be seen from Fig. 6(d) that these sources, seemingly related to muscle activity, are, in fact, the mixture of brain and muscle activity, given that only the last four sources were removed during reconstruction according to the criterion shown in Fig. 4. Moreover, the 15th source (the 5th from the last) also contains both brain and muscle activity, quite obviously from 8 to 10 s. By CCA, the SOS seems unable to sufficiently separate the underlying sources, since even the distinct-independent muscle activities are mixed with each other. Furthermore, the selected removable muscle-activity sources in both ICA and CCA involve valuable EEG signals, seeing the fluctuation in the segments, which should have been flat. Compared with ICA and CCA, IVA almost overcomes all the above-mentioned problems, leading to the most ultimate source separation. The last four sources are just the same as the original four muscle activities, and meanwhile there is almost no mutual mixing situation between brain and muscle activity. This can also be further verified by examining the cleaned EEG data after reconstruction, as shown in Fig. 6(d)–(f). Note that Fig. 6(d)–(f) presents the amplified version of the data for easily and clearly observing the signal details. We can see that the original EEG signals, which have been heavily contaminated, were recovered much better by IVA than by ICA and CCA. Among all the three methods, the recovered

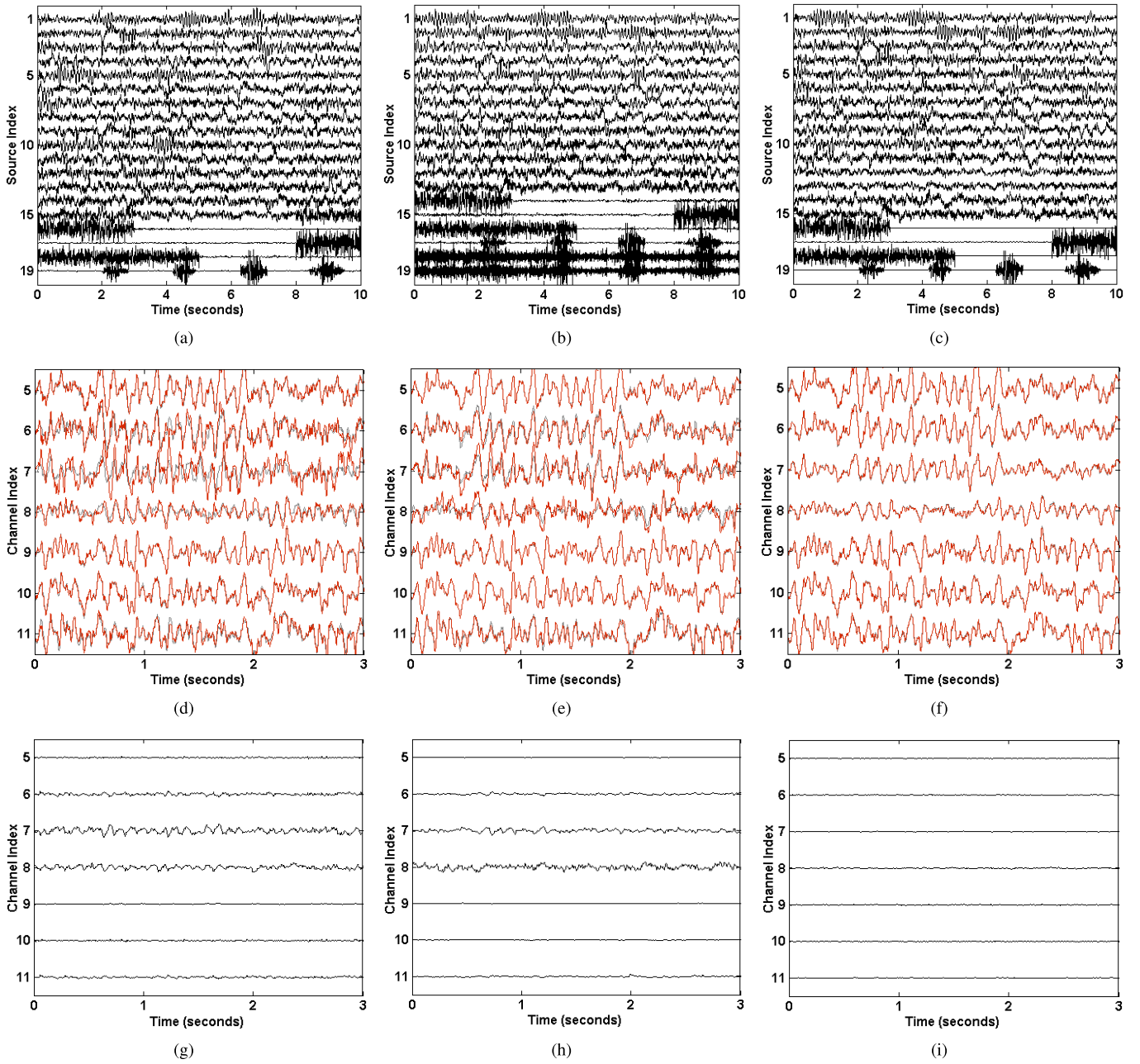


Fig. 6. Some decomposition and reconstruction results for one EEG data set heavily contaminated by muscle activity at SNR = 0.5. The extracted underlying sources using (a) ICA, (b) CCA, and (c) IVA. The amplified display of the cleaned EEG signals (red curves) after using (d) ICA, (e) CCA, and (f) IVA in contrast to the original pure EEG (gray curves). The error between the original pure EEG and the cleaned EEG calculated by (g) corresponding to (g) ICA, (h) CCA, and (i) IVA.

EEG signals by using ICA were distorted the most, although the decomposition results of ICA were seemingly promising. This indicates the fact that the last four sources extracted by ICA actually contain valuable brain activity and thus ICA is unable to effectively isolate muscle artifacts. The performance of CCA seems a little better than that of ICA without that much distortion, but obvious muscle activity retained (e.g., the eighth channel). By using IVA, muscle artifacts have been more effectively removed, while brain activity has been better preserved. To further clarify the performance of the three methods, we calculated the error for each channel between the original pure EEG and the three cleaned EEG signals,

respectively, according to the following equation:

$$\begin{aligned} \text{ERROR}(p, t) &= (\mathbf{X}_{\text{EEG}}(p, t) - \tilde{\mathbf{X}}_{\text{EEG}}(p, t)) / \max(\text{abs}(\mathbf{X}_{\text{EEG}}(p, :))) \end{aligned} \quad (9)$$

with p , t , $\max$, and abs representing the channel index, the time index, the maximal value, and the absolute value of a time series. The corresponding plots for these errors have been shown in Fig. 6(g)–(i). We can see that IVA leads to the least error and distortion for the original EEG compared with ICA and CCA.

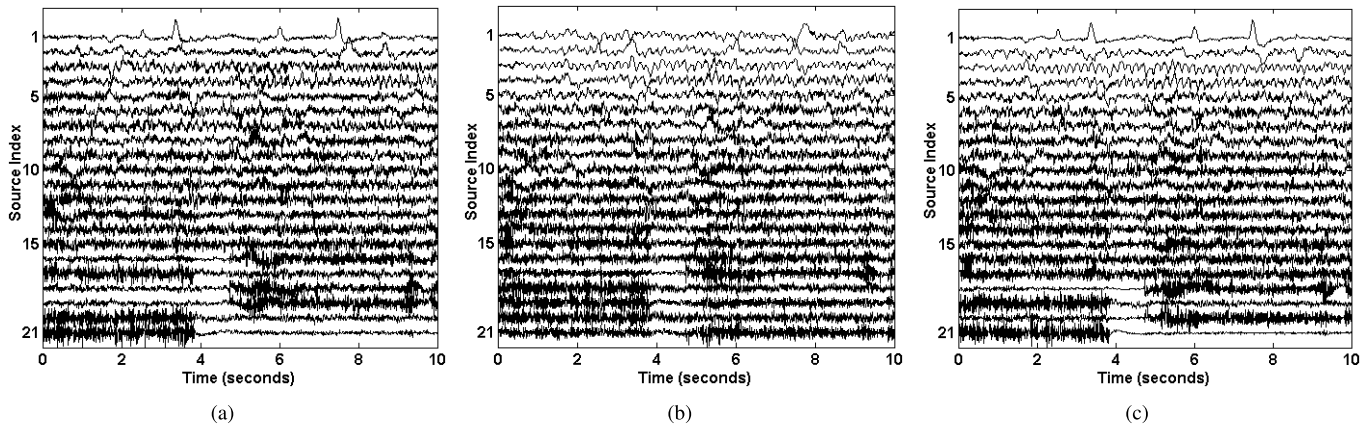


Fig. 7. Decomposition results for the 10-s ictal EEG heavily contaminated by muscle activity. The extracted underlying sources using (a) ICA, (b) CCA, and (c) IVA.

B. Results of Real Data

In this section, we examined the practical performance of the three methods, when applied to the real ictal EEG data. Since the ground truth was unknown, the performance judgment basically depended on visual inspection. It is worth mentioning that this ictal EEG data set was contaminated with severe muscle artifacts and particularly challenging. To the best of our knowledge, in the literature, there exist no effective multichannel BSS techniques for handling this challenging data set. Even the researchers specializing in using multichannel ICA and CCA for ictal EEG denoising have rarely reported the results for this data set in the related work [20], [23]–[25], [39]. Therefore, we do not expect that IVA will perfectly remove muscle artifacts for this data, but do comparatively analyze the decomposition results of the three methods, demonstrating the possible merits of IVA in contrast to ICA and CCA.

The extracted underlying sources by applying the three methods to the contaminated ictal data are shown in Fig. 7(a)–(c). Analogous to the simulation results, IVA can better isolate the major muscle activities into the last several sources, nevertheless, muscle activities in the ICA and CCA decomposition are spread more widely across multiple sources. In particular, IVA can better recover the transient muscle activity than ICA and CCA, seeing the relatively flatter segments that should contain minimal brain and muscle activity. For example, the last source in Fig. 7(c), representing a transient muscle activity from 0 to 3.8 s, possesses the least impurity between 3.8 and 10 s, while the last ones in Fig. 7(a) and (b) contain relatively more ingredients between 3.8 and 10 s. The similar results can also be obtained by comparatively observing the fourth source in Fig. 7(a), the fifth one in Fig. 7(b), and the fourth one in Fig. 7(c).

Another interesting point lies in the first several sources extracted by the three methods. ICA can effectively isolate the eye-blink activity into its first source [Fig. 7(a)], while CCA seems ineffective for this type of activity [Fig. 7(b)]. However, CCA is able to perfectly recover the ictal activity [e.g., the third source in Fig. 7(b)], while the same counterpart in ICA [e.g., the third in Fig. 7(a)] is apparently the mixture

of ictal activity and muscle artifact. These findings are consistent with previous studies supporting the usage of ICA for eye-blink activity removal and CCA for muscle-activity removal [22], [25]. When IVA was applied, both types of activities were extracted effectively [e.g., the first and the third sources in Fig. 7(c)]. It can be seen from Fig. 7(c) that IVA can better isolate the obvious transient muscle activity into the last several sources, perfectly recover the eye-blink activity into the first source, and meanwhile exactly extract the sources related to ictal activity. This is just the integration of the advantages of both ICA and CCA.

IV. DISCUSSION AND CONCLUSION

Numerous previous studies have demonstrated that ICA and CCA are useful methods for muscle artifact removal in EEG data [14]–[16], [20]–[25]. In the literature, some researchers support the usage of CCA [20]–[22], while many others prefer ICA [14], [16]. However, it is well known that muscle artifacts often involve the movement of a group of muscles and thus do not possess a stereotyped topography [21]. In this case, ICA do not function well, leading to the consequence that too much brain activity has to be sacrificed for sufficiently removing muscle artifacts. In addition, by exploiting only the second-order statistical information, CCA is unable to thoroughly isolate muscle activity. As recently surveyed in [15], which method performs better may highly depend on the type of the EEG signal. Here, it is worth emphasizing that the goal of this paper is not to discuss which method is better than the other, but to consider their complementary properties.

In this paper, we propose a more effective IVA-based method to remove muscle artifacts from EEG data by naturally integrating the HOS and SOS into one single BSS framework. In such a framework, the sources with maximal independence and meanwhile maximal autocorrelation can be eventually extracted. This is just the combination of the merits of both ICA and CCA-based methods. A more theoretical explanation may be given in the aspect of statistical diversity. The goal of BSS can be achieved using different types of statistical diversity, including SOS, HOS, sample dependence, noncircularity, nonstationarity, and statistical dependence across multiple data

sets. As we have mentioned before, ICA generally exploits HOS for identifying statistically independent sources; CCA utilizes the second-order statistical dependence across two data sets to find the maximally self-autocorrelated sources subsequently, and meanwhile, these sources are statistically uncorrelated; IVA employs both HOS and the second-order statistical dependence across multiple data sets for jointly achieving independent decomposition of multiple data sets. It has been shown that IVA can be superior to identifying sources in each data set independently using an ICA algorithm followed by an optimal source alignment algorithm. In addition, IVA can achieve more accurate source separation than CCA [28]. A recent study [42] also reported that, with the addition of each new statistical diversity type, identification of a broader class of signals becomes possible. The author presented a number of examples from various domains to demonstrate that, in many scenarios, it might be worthwhile to jointly account for multiple statistical diversities. This is also the case in our EEG denoising application. From many previous studies and a recent comprehensive review [15], it is clear that individual muscle activity simultaneously possesses two important characteristics: statistical independence and low autocorrelation. In the past, researchers separately utilized the two characteristics for muscle artifact removal. However, since the number of muscles in the head, face, and neck is large, the measured muscle activity is often generated from a group of muscles and mixed up. In this case, only exploiting HOS is sometimes insufficient to achieve ultimate muscle artifact isolation, especially with a limited number of channels. Thus, considering additional low-autocorrelation characteristic is quite necessary. Under this situation, IVA should be an appropriate candidate. We have demonstrated using realistic simulated data and real data analysis that, when the contamination is heavy with low SNR, IVA is able to separate muscle and brain activities effectively. In contrast, neither ICA nor CCA can function well in such a situation. Even when the EEG data are slightly or moderately contaminated by muscle activity, IVA is superior to both ICA and CCA. Our results suggest that IVA performs consistently better, irrespective of the contamination level in the EEG data.

We also fortuitously found that the eye-blink activity could be effectively extracted in the real data analysis, possibly because HOS has been utilized in IVA. This provides a possible chance to simultaneously remove the artifacts related to eye and muscle activities in one single step, instead of dealing with these two types of artifacts using ICA and CCA separately as currently do in the literature [22]. Nevertheless, this needs further comprehensive verification in the future. In addition, as we mentioned in Section I, IVA can be applied to analyze other types of signals in the I&M field as long as the decomposition needs the integration of the two different statistical diversities, HOS and SOS. The most possible usage may include the isolation of white noise, transient disturbance, motion artifact, and so on.

It should be also noted that IVA can be treated as the main backbone of the methods based on multiple processing stages. According to our practical experience, the optimal method for artifact removal may be composed of combining more than one

technique to sufficiently correct the signal through a multistage strategy. In the I&M field, the single-channel techniques, such as wavelet transform [17], [29], empirical mode decomposition [30], [31], least mean squares [32], [33], morphological component analysis [40], [41], and so on, have been widely used for signal denoising. Meanwhile, the combination of the single-channel techniques and the multichannel ICA and CCA methods has also been quite popular [10], [17], [30]. Therefore, it is inspired that those single-channel techniques can be exploited prior to and/or posterior to the IVA method, forming multistage-based single-channel or multichannel artifact removal methods. In particular, by using morphological component analysis, EEG signals in each channel can be decomposed into components that have different morphological characteristics. Each component is sparsely represented by different bases, e.g., discrete cosine transform, wavelet, and Dirac basis. In this case, the contaminated EEG signals may be better explained due to the use of several different morphological waveforms in an overcomplete dictionary. When combined with IVA as a postprocessing step, this may allow much finer signal correction and performance improvement. This is an option largely unexplored by researchers in the area. However, this is beyond the scope of this paper and will be the subject of further research.

We tested the computational time cost of the three methods over the 10-s 19-channel EEG through 100-independent implementations. The implementation was done in the MATLAB (MathWorks Inc., Novi, MI, USA) and run under 64-b Microsoft Windows 8 OS on a computer with Dual Intel(R) Core(TM) i5-3427U 1.80-GHz CPU and 8.00-GB RAM. The mean computational time for ICA was 1.1249 s with a standard deviation 0.2581; the mean for CCA was 0.0158 s with a standard deviation 0.00024; the mean for IVA was 59.8544 s with a standard deviation 0.5003. We can see that IVA consumes much more time than ICA and CCA. This limits the IVA-based method to be only used under the offline situation instead of online.

Moreover, although the IVA-based method has been examined to better handling the problem in this paper, whether it is applicable for all situations (e.g., EEG in other physical or mental diseases) still needs to be extensively investigated. We believe this paper at least provides a novel choice for neuroscientists and clinicians, can encourage them to verify this tool in a variety of real clinical settings, and may benefit various EEG-based practical applications.

REFERENCES

- [1] S.-F. Liang, Y.-H. Kuo, Y.-H. Hu, Y.-H. Pan, and Y.-H. Wang, "Automatic stage scoring of single-channel sleep EEG by using multiscale entropy and autoregressive models," *IEEE Trans. Instrum. Meas.*, vol. 61, no. 6, pp. 1649–1657, Jun. 2012.
- [2] A. Lay-Ekuakille *et al.*, "Entropy index in quantitative EEG measurement for diagnosis accuracy," *IEEE Trans. Instrum. Meas.*, vol. 63, no. 6, pp. 1440–1450, Jun. 2014.
- [3] S. Siuly, H. Wang, and Y. Zhang, "Detection of motor imagery EEG signals employing Naïve Bayes based learning process," *Measurement*, vol. 86, pp. 148–158, May 2016.
- [4] Y. Dai, X. Wang, X. Li, and Y. Tan, "Sparse EEG compressive sensing for Web-enabled person identification," *Measurement*, vol. 74, pp. 11–20, Oct. 2015.

- [5] X. Chen, A. Liu, M. J. McKeown, H. Poizner, and Z. J. Wang, "An EEMD-IVA framework for concurrent multidimensional EEG and unidimensional kinematic data analysis," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 7, pp. 2187–2198, Jul. 2014.
- [6] Y.-C. Chen, B.-S. Lin, and J.-S. Pan, "Novel noncontact dry electrode with adaptive mechanical design for measuring EEG in a hairy site," *IEEE Trans. Instrum. Meas.*, vol. 64, no. 12, pp. 3361–3368, Dec. 2015.
- [7] L.-D. Liao *et al.*, "A novel 16-channel wireless system for electroencephalography measurements with dry spring-loaded sensors," *IEEE Trans. Instrum. Meas.*, vol. 63, no. 6, pp. 1545–1555, Jun. 2014.
- [8] E. Urrestarazu, J. Iriarte, M. Alegre, M. Valencia, C. Viteri, and J. Artieda, "Independent component analysis removing artifacts in ictal recordings," *Epilepsia*, vol. 45, no. 9, pp. 1071–1078, 2004.
- [9] B. W. McMenamin, A. J. Shackman, L. L. Greischar, and R. J. Davidson, "Electromyogenic artifacts and electroencephalographic inferences revisited," *NeuroImage*, vol. 54, no. 1, pp. 4–9, 2011.
- [10] B. Gao, L. Bai, W. L. Woo, G. Y. Tian, and Y. Cheng, "Automatic defect identification of eddy current pulsed thermography using single channel blind source separation," *IEEE Trans. Instrum. Meas.*, vol. 63, no. 4, pp. 913–922, Apr. 2014.
- [11] C.-Y. Yu, Y. Li, B. Fei, and W.-L. Li, "Blind source separation based X-ray image denoising from an image sequence," *Rev. Sci. Instrum.*, vol. 86, no. 9, p. 093701, 2015.
- [12] P. Mishra and S. K. Singla, "Artifact removal from biosignal using fixed point ICA algorithm for pre-processing in biometric recognition," *Meas. Sci. Rev.*, vol. 13, no. 1, pp. 7–11, 2013.
- [13] A. M. F. L. M. de Sá, J. M. de Seixas, J. D. C. Junior, D. D. Ferreira, and A. S. Cerqueira, "A principal component-based algorithm for denoising in single channel data (PCA for denoising in single channel data)," *Measurement*, vol. 60, pp. 121–128, Jan. 2015.
- [14] L. Albera *et al.*, "ICA-based EEG denoising: A comparative analysis of fifteen methods," *Bull. Polish Acad. Sci., Tech. Sci.*, vol. 60, no. 3, pp. 407–418, 2012.
- [15] J. A. Urigüen and B. Garcia-Zapirain, "EEG artifact removal—State-of-the-art and guidelines," *J. Neural Eng.*, vol. 12, no. 3, p. 031001, 2015.
- [16] I. Winkler, S. Brandl, F. Horn, E. Waldburger, C. Allefeld, and M. Tangermann, "Robust artifactual independent component classification for BCI practitioners," *J. Neural Eng.*, vol. 11, no. 3, p. 035013, 2014.
- [17] N. Mammone, F. La Foresta, and F. C. Morabito, "Automatic artifact rejection from multichannel scalp EEG by wavelet ICA," *IEEE Sensors J.*, vol. 12, no. 3, pp. 533–542, Mar. 2012.
- [18] A. J. Shackman, B. W. McMenamin, H. A. Slagter, J. S. Maxwell, L. L. Greischar, and R. J. Davidson, "Electromyogenic artifacts and electroencephalographic inferences," *Brain Topogr.*, vol. 22, no. 1, pp. 7–12, 2009.
- [19] H. Nam, T.-G. Yim, S. K. Han, J.-B. Oh, and S. K. Lee, "Independent component analysis of ictal EEG in medial temporal lobe epilepsy," *Epilepsia*, vol. 43, no. 2, pp. 160–164, 2002.
- [20] W. De Clercq, A. Vergult, B. Vanrumste, W. Van Paesschen, and S. Van Huffel, "Canonical correlation analysis applied to remove muscle artifacts from the electroencephalogram," *IEEE Trans. Biomed. Eng.*, vol. 53, no. 12, pp. 2583–2587, Nov. 2006.
- [21] J. Gao, C. Zheng, and P. Wang, "Online removal of muscle artifact from electroencephalogram signals based on canonical correlation analysis," *Clin. EEG Neurosci.*, vol. 41, no. 1, pp. 53–59, 2010.
- [22] M. R. Mowla, S.-C. Ng, M. S. A. Zilany, and R. Paramesran, "Artifacts-matched blind source separation and wavelet transform for multichannel EEG denoising," *Biomed. Signal Process. Control*, vol. 22, pp. 111–118, Sep. 2015.
- [23] A. Vergult *et al.*, "Improving the interpretation of ictal scalp EEG: BSS-CCA algorithm for muscle artifact removal," *Epilepsia*, vol. 48, no. 5, pp. 950–958, 2007.
- [24] M. De Vos *et al.*, "Removal of muscle artifacts from EEG recordings of spoken language production," *Neuroinformatics*, vol. 8, no. 2, pp. 135–150, 2010.
- [25] H. Hallez *et al.*, "Removing muscle and eye artifacts using blind source separation techniques in ictal EEG source imaging," *Clin. Neurophysiol.*, vol. 120, no. 7, pp. 1262–1272, 2009.
- [26] X. Chen, A. Liu, H. Peng, and R. K. Ward, "A preliminary study of muscular artifact cancellation in single-channel EEG," *Sensors*, vol. 14, no. 10, pp. 18370–18389, 2014.
- [27] X. Chen, A. Liu, J. Chiang, Z. J. Wang, M. J. McKeown, and R. K. Ward, "Removing muscle artifacts from EEG data: Multichannel or single-channel techniques?" *IEEE Sensors J.*, vol. 16, no. 7, pp. 1986–1997, Apr. 2016.
- [28] M. Anderson, T. Adalı, and X.-L. Li, "Joint blind source separation with multivariate Gaussian model: Algorithms and performance analysis," *IEEE Trans. Signal Process.*, vol. 60, no. 4, pp. 1672–1683, Apr. 2012.
- [29] C.-C. Liao, H.-T. Yang, and H.-H. Chang, "Denoising techniques with a spatial noise-suppression method for wavelet-based power quality monitoring," *IEEE Trans. Instrum. Meas.*, vol. 60, no. 6, pp. 1986–1996, Jun. 2011.
- [30] Y.-W. Tang, C.-C. Tai, C.-C. Su, C.-Y. Chen, and J.-F. Chen, "A correlated empirical mode decomposition method for partial discharge signal denoising," *Meas. Sci. Technol.*, vol. 21, no. 8, p. 085106, 2010.
- [31] R. Hussein, K. B. Shaban, and A. H. El-Hag, "Wavelet transform with histogram-based threshold estimation for online partial discharge signal denoising," *IEEE Trans. Instrum. Meas.*, vol. 64, no. 12, pp. 3601–3614, Dec. 2015.
- [32] M. R. Ram, K. V. Madhav, E. H. Krishna, N. R. Komalla, and K. A. Reddy, "A novel approach for motion artifact reduction in PPG signals based on AS-LMS adaptive filter," *IEEE Trans. Instrum. Meas.*, vol. 61, no. 5, pp. 1445–1457, May 2012.
- [33] I. Soumya, K. Z. U. Rahman, D. V. R. K. Reddy, and A. Lay-Ekuakille, "Efficient block processing of long duration biotelemetry brain data for health care monitoring," *Rev. Sci. Instrum.*, vol. 86, no. 3, p. 035003, 2015.
- [34] A. Belouchrani, K. Abed-Meraim, J.-F. Cardoso, and E. Moulines, "A blind source separation technique using second-order statistics," *IEEE Trans. Signal Process.*, vol. 45, no. 2, pp. 434–444, Feb. 1997.
- [35] J.-F. Cardoso, "High-order contrasts for independent component analysis," *Neural Comput.*, vol. 11, no. 1, pp. 157–192, Jan. 1999.
- [36] O. Friman, M. Borga, P. Lundberg, and H. Knutsson, "Exploratory fMRI analysis by autocorrelation maximization," *NeuroImage*, vol. 16, no. 2, pp. 454–464, 2002.
- [37] T. Kim, T. Eltoft, and T.-W. Lee, "Independent vector analysis: An extension of ICA to multivariate components," in *Independent Component Analysis and Blind Signal Separation*. Heidelberg, Germany: Springer, 2006, pp. 165–172.
- [38] J. Laney, K. P. Westlake, S. Ma, E. Woytowicz, V. D. Calhoun, and T. Adalı, "Capturing subject variability in fMRI data: A graph-theoretical analysis of GICA vs. IVA," *J. Neurosci. Methods*, vol. 247, pp. 32–40, May 2015.
- [39] M. De Vos *et al.*, "Automated artifact removal as preprocessing refines neonatal seizure detection," *Clin. Neurophysiol.*, vol. 122, no. 12, pp. 2345–2354, 2011.
- [40] J.-L. Starck, Y. Moudden, J. Bobin, M. Elad, and D. L. Donoho, "Morphological component analysis," *Proc. SPIE*, vol. 5914, p. 59140Q, May 2005, doi: 10.1117/12.615237.
- [41] X. Yong, R. K. Ward, and G. E. Birch, "Artifact removal in EEG using morphological component analysis," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Apr. 2009, pp. 345–348.
- [42] T. Adalı, M. Anderson, and G.-S. Fu, "Diversity in independent component and vector analyses: Identifiability, algorithms, and applications in medical imaging," *IEEE Signal Process. Mag.*, vol. 31, no. 3, pp. 18–33, May 2014.

Xun Chen, photograph and biography not available at the time of publication.

Hu Peng, photograph and biography not available at the time of publication.

Fengqiong Yu, photograph and biography not available at the time of publication.

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